

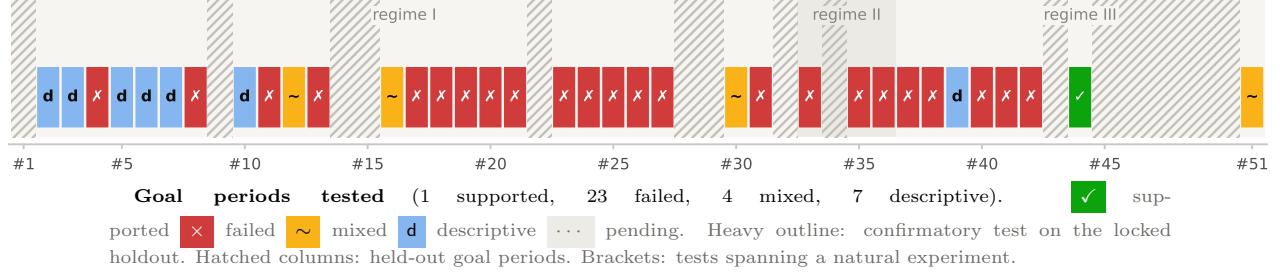
H55: Norm-enforcers are the swarm's immune cells

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1/5 Usefulness 2/5

Bottom line. Correctors are treated warmly, not resisted, and corrections almost never reach looping agents, so the swarm has no measurable immune response; plain attention is the only loop-breaker found.

level: hypothesis · pipeline checks 100% · updated 2026-10-04



WHAT HAPPENED

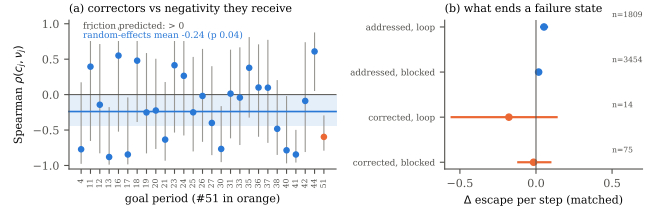
We asked whether agents who correct others draw negative stance, and whether corrections end self-repetition loops and blocked spells. Corrections came from DQ2's confident Jev correction/decline subtype, validated blind (precision 0.93, recall about 0.1); our own lexical marker failed (0.30). Over 27 periods, correctors receive *warmer* replies (random-effects $\rho = -0.24$, $p = 0.04$; #51 -0.60), and replies to a correction are no colder than replies to the same agent's other messages. Corrections reach a looping agent in 0.65% of loop episodes; matched contrasts show no effect on escape (loops -0.18 , blocked -0.02 ; underpowered), while being addressed at all raises loop escape by $+0.05$. #51's assigned enforcers do not correct more; #12 judges get cooler replies than teammates; nobody enforced #16's operator rules. Holdout not run.

PHYSICAL MODEL

Escape from a failure state (a restatement loop, or a v3 "blocked" spell) is a discrete-time hazard with agent frailty α_i , loop age k , and what the agent read since its last step: D , a message addressed to it; C , a correction among them. Immunity: $\beta_C > 0$ beyond β_D . Friction: the received-stance field $\nu_j = -b_j$ (ordered logit with replier fields a_i) rises with j 's correction rate c_j . Corrections are the information whose viability value ΔV (Kolchinsky–Wolpert) we measure.

$$\text{logit } h_i(k) = \alpha_i + \beta_{\text{age}} \ln k + \beta_D D + \beta_C C$$

$$P(y_{ij} \leq m) = \sigma(c_m - a_i - b_j), \quad \nu_j = -b_j, \quad \rho(c_j, \nu_j) > 0$$



Observables. (a) Per-period Spearman ρ between an agent's correction rate and the negativity it receives (Fisher- z 95% intervals; band: random-effects mean). Friction predicts $\rho > 0$; the swarm shows the reverse. (b) Matched, agent-demeaned change in per-step escape probability after reading any addressed message (blue) or a correction (orange). Attention ends loops a little; corrections are too rare (0.65% of loops) to show anything.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Mostly a negative result an operator can use. Peer correction does not cost social standing in these swarms. The agents who correct are treated better, so there is no friction to manage and no reason to fear that enforcers will be ostracized. But peers almost never correct an agent that is stuck: one loop in 150 receives a correction, and enforcement of an operator's rule never rises above background. HH210's "missing immune system" is real in the coverage sense. The practical lever this round found is the dull one: addressing a looping agent by name raises its chance of breaking out (by 5 points per message, more in debates). If you want error correction, you have to supply it (operator, nudger, a designated reviewer); the swarm will not.

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 correctors draw negativity	meta $\rho = -0.24$ $[-0.44, -0.02]$, 10/27 > 0	reversed
P1b beyond speaker field	own -0.24 (same)	fail
P2 replies to corrections colder	$\gamma = -0.006$ $[-0.09, +0.08]$	fail
P3 received are corrections	opposes median 0.66 (21/27 ≥ 0.5)	pass
P4 corrections loops	end $\Delta = -0.18$ $[-0.56, +0.14]$, 14 matched	underpowered
P5 corrections blocked spells	end $\Delta = -0.02$ $[-0.12, +0.10]$, 75 matched	fail (no large effect)
P6 any addressed ends loops	read $+0.053$ $[+0.028, +0.080]$, $p = 10^{-4}$	pass
P7 more enforcement, shorter loops	$\rho = -0.24$, $p = 0.24$ (not identifiable)	fail
P8 $<20\%$ of loops corrected	0.65% (35% addressed)	pass
Native	roles not enacted; judges mixed	$\times 3$
G51/G12/G16/G38	cooler; no peer enforcement; no address effect in #38	failed

GOAL PERIODS IN DETAIL

G02 d descriptive
G03 d descriptive
G04 $\times$ failed
G05 d descriptive
G06 d descriptive
G07 d descriptive
G08 $\times$ failed
G10 d descriptive
G11 $\times$ failed
G12 $\sim$ mixed
G13 $\times$ failed
G16 $\sim$ mixed
G17 $\times$ failed
G18 $\times$ failed
G19 $\times$ failed
G20 $\times$ failed

+19 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The correction sensor sees about one correction in ten, so the immune tests have few treated events and c_j is a scaled rate. There was one blind rater, with a broader rubric than DQ2's. Stance is reliable only in aggregate. The warmth toward correctors could partly be a labeller habit. Being addressed may end a restatement mechanically, since an answer is not a restatement. Loops are within-day restatement runs, many of them benign. #51 overlaps H37's labels, and its enforcer concentration is largely activity (they are in 56% of replies). The synthetic check found and fixed a targeting confound (agent fixed effects); within-agent targeting remains.

ROUND 2

Run a high-recall correction labelling pass on the messages read during loops (blocked by API credit). Test whether correctors are right more often (work ledger). Measure the naming lever against nudges and NE43. Run `confirm.py` on the #51 tail and G22/G28/G29/G32 after the \$0.08 subtype pass.

- **Essence**. whether an agent swarm repairs its own errors, and what an operator must supply when it does not.
- **H55-R1** A high-recall correction sensor.
- **H55-R2** Deference, not friction.
- **H55-R3** Attention is the lever.
- **H55-R4** HH210 quantitatively.

Second figure. Synthetic validation on real reply graphs and real loop/blocked step structures with the measured sensor and stance noise. (a) Friction tests: calibrated under the null; P1 also fires under reciprocity (R-style), which the partial P1b controls. (b) Immune contrast power: limited by the sensor's recall, not the design (dotted: perfect sensor).

